A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1 LEO:** high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover **1** But each area is only revisited every 15 days so they cannot follow rapidly changing weather **1 Geostationary:** constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
02.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Wind velocity **1** Ocean currents / sea surface temperature **1** Wave height **1** Ice thickness **1** Vegetation cover **1** Sea level **1** Atmospheric temperature / composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome **1** If the prediction matches, the model can be trusted (with caution) – it is continually modified as more data are collected and differences analysed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Models need data from all regions / oceans / altitudes – gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions **1** Unrepresentative data (eg urban heat islands, short time series) would bias the model – remote sensing helps fill gaps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gravimetry:** gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) **1** Magnetometry **1 Seismic survey:** sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata **1** Resistivity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Trial drilling is the most expensive technique per site **1** Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites **1** Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Different minerals emit / reflect infrared at different wavelengths **1** Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
03.4	1 mark for the point below. **1** Results are available immediately in the field, saving time waiting for laboratory analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (6400 - 220) / 6400 x 100 **1** = 96.6% (accept 97%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO2 source **1** Flue gas desulfurisation fitted to remaining coal plants – coal / oil desulfurisation **1** NOx comes mainly from road transport and vehicle numbers / distance travelled increased greatly **1** NOx is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur **1** Catalytic converters only work when warm / diesel engines produce more NOx / older vehicles remained in use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO2 (broken down by UV to monatomic oxygen which forms ozone) **1** If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Lichens are very sensitive to acidic conditions / SO2 and different species have different tolerances **1** The species present, their abundance, size / health can be used as a biotic index of the pollution level **1** They give a long-term integrated measure of pollution rather than a single instantaneous reading – cheap, no equipment needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) **1** Most phosphorus compounds have low solubility so little is available in soil water **1** Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil **1** Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake • **symbiotic** → a close relationship in which both species benefit	2
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) **1** Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment **1** Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates) – harvesting removes phosphorus from farmland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock **1** Breed crops that absorb phosphate more efficiently **1** Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) **1** Recover phosphate from sewage effluent by iron(III) sulfate precipitation – reduce soil erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (2900 - 1400) / 1400 x 100 **1** = 107% (accept 107.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Large decrease from 24 per year (1975-79) to 2 per year (2010-14) **1** Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Double-hulled tankers required for all large tankers **1** Improved navigation (GPS, AIS) and offshore shipping routes **1** Inert gas systems reducing explosions **1** Twin engines / rudders reducing groundings after mechanical failure **1** Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) **1** Separate ballast tanks / crude oil washing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff **1 Routine operations:** small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) **1** Growth in oil use and transport means more small releases even if large accidents are rarer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **diffuse** → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 5 points below. **1 Terracing:** sloping land divided into narrow level fields held by retaining walls along contours – water slows as it crosses each terrace and infiltrates instead of running off **1 Tied ridging:** on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff **1 Windbreaks:** hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles – protect bare fields after harvest **1 Mulching:** organic matter on the surface protects soil from raindrop impact and wind – decomposes to humus which binds particles – retains moisture **1 Strip cropping:** strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field – slows runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Terracing** → cutting the slope into level steps • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	5
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles **1** Permanent root systems and canopy cover protect the soil all year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Control stocking density so vegetation regrows faster than it is eaten **1** Restrict access to vulnerable areas such as river banks (fencing) **1** Move / remove livestock at high-risk times, eg after heavy rain when soil is wet **1** Rotational grazing to rest pastures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Chi-squared test (frequencies / counts in categories) **1** t-test (two means of measured data) **1** Spearman's rank correlation coefficient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Chi-squared** → a test comparing observed frequencies with expected frequencies **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4
08.2	1 mark for the point below. **1** Chi-squared compares frequencies / counts in categories – it cannot be used for continuously variable measured data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **chi-squared** → a test comparing observed frequencies with expected frequencies **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	1
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level **1** The negative sign shows a negative correlation: one variable decreases as the other increases **1** There is less than a 5% probability the correlation arose by chance – reject the null hypothesis (but correlation is not proof of causation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	3

Do not write outside the box

Question	Marking guidance	Marks
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Show the standard deviation / range of the original values around each mean **1** If the bars of two means overlap greatly the difference is probably not significant – little overlap suggests a significant difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
09.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
09.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = £11 million **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
10.2	**1 mark each** 1 mark for each of the 3 points below. **1** It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided **1 ALARA:** emissions should be As Low As Reasonably Achievable **1 BATNEEC:** the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** As industries expand or cities grow the number of pollutant sources increases **1** So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Critical Pathway Analysis:** predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk – mainly radioactive discharges – checked by sampling • **Critical Group Monitoring:** exposure of the most-exposed group monitored before health effects – if their risk is acceptable everyone else is safer – emissions reduced if not • **Emission control strategies:** location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • **Principles:** polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • **Selection of methods:** production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • **Evaluation:** prevention usually cheaper and more effective than remediation – costs rise steeply near 100% – CPA relies on understanding of pathways – enforcement and international agreement needed for transboundary pollutants – judgement about acceptable risk is political as well as scientific • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **Critical Group** → the people most at risk / most heavily exposed **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• The Level 3 answer explains *how* each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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